ADITYA UNIVERSITY

Department of Computer Science and Engineering (MI)
Machine Learning Corner Stone Project
Batch: 3
Title : Smart Agriculture Prediction System

Abstract

The Smart Agriculture Prediction System is designed to assist farmers in making informed decisions using Machine Learning and data analytics. Agriculture is affected by several factors such as soil conditions, weather variations, water availability, and crop diseases. The proposed system analyzes important agricultural parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, soil pH, and rainfall to recommend the most suitable crop for cultivation.

The system uses agricultural datasets to preprocess and analyze the required features and trains a Machine Learning model for crop recommendation. A Flask-based backend is used to connect the trained Machine Learning model with a user-friendly web interface. Users can enter the required soil and environmental parameters, and the system processes the input and provides a suitable crop recommendation.

The proposed system can be further extended with additional features such as real-time weather forecasting, smart irrigation recommendations, fertilizer suggestions, and plant disease detection using image processing. The system aims to improve agricultural productivity, reduce resource wastage, minimize farming risks, and support sustainable farming practices through data-driven decision making.

Keywords: Machine Learning, Smart Agriculture, Crop Recommendation, Data Analytics, Flask, Soil Analysis, Weather Data, Precision Agriculture

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